

Hackathon 4

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The quantum dots are man-made nanostructures that behave like atoms especially in terms of discrete energy levels and distribution of the electrons. These nanostructures have found a wide range of applications. A particluar interesting application is the use in the flash memories also called as nanocrystal flash memories by Sandip Tiwari and his team at IBM in 1990s (Sandip Tiwari et al APL 1996). These nanostructures (nanocrystals) are placed in the oxides of MOS structures for creating specific energy levels for trapping. The number of electrons stored in these traps can be modulated by applying bias on the metal side. These stored electrons further modulate the conductivity of the underlying MOSFET by Coulombic forces. Once these electrons are trapped it is unwanted that they come out of it by themselves as this leads of corruption or loss of the data stored in the memory. The amount of time needed these electrons stay in these nanocrystals is called as the retention time. The deeper the trap energy more time it would require to escape from the trap and hence better retention. For this case the band diagram and the physical structure is as shown. In a band diagram shown on the board the reference energy level ($E=0$) in this case is the energy at which the electron is completely free from the underlying crystal also called as the vacuum level. Calculate the least amount of energy that must be supplied an electron stored in the nanocrystal to become free from the nanocrystal and the energy of the electron that is stored in the nanocrystal using open-ended method. For the purpose of this hackathon, ignore the interaction between the electrons and neighbouring quantum wells

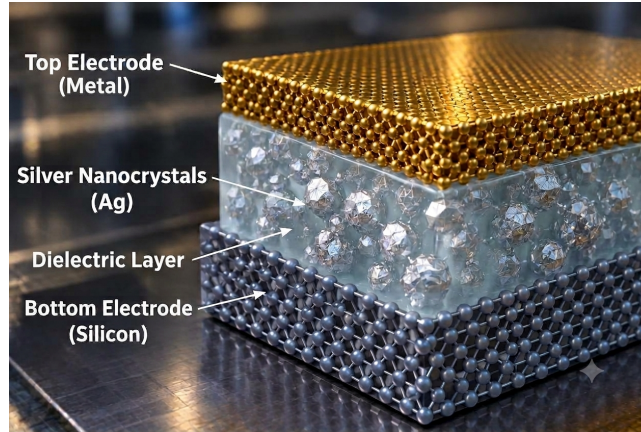


Figure 1: AI Generated Representational Image of nanocrystal Flash memory

Given: For a Finite Energy potential well with a well width L (as discussed in SDF) the Eigen value equation is given by

$$\begin{aligned}\frac{2k\kappa}{k^2 - \kappa^2} &= \tan(kL) \\ k &= \frac{\sqrt{2mE}}{\hbar} \\ \kappa &= \frac{\sqrt{2m(U_0 - E)}}{\hbar}\end{aligned}$$

The potential energy of the Finite well for the above E-k relationship is

$$\begin{aligned}U(x) &= 0 & 0 < x < a \\ U(x) &= 1 & 0 > x \text{ and } x > a\end{aligned}$$